MSE 311 Thermodynamics of materials-Homework 11

Name: *Student ID:*

1. Does the liquid atom have coordination numbers? Please show how to calculate it.

2. Prove the following equation when $\eta \neq 0$:

$$\Delta S^{M,id} > \Delta S^{M}, \text{ where } \Delta S^{M} = -\frac{R}{2}\left(\begin{aligned}&\left(X_A+\frac{\eta}{2}\right)\ln\left(X_A+\frac{\eta}{2}\right)+\left(X_B+\frac{\eta}{2}\right)\ln\left(X_B+\frac{\eta}{2}\right)\\&+\left(X_A-\frac{\eta}{2}\right)\ln\left(X_A-\frac{\eta}{2}\right)+\left(X_B-\frac{\eta}{2}\right)\ln\left(X_B-\frac{\eta}{2}\right)\end{aligned}\right)$$

3. Simplification the (ΔG^{XS}) in this condition

$$\Delta G^{XS} = \Delta H^{M} - T\left(\Delta S^{M} - \Delta S^{M,id}\right)$$

where, $\Delta H^{M} = \Omega_{AB}\left(X_A X_B + \frac{\eta^2}{4}\right)$

4. Copper and gold form the complete range of solid solution at temperatures between 410°C and 889°C, and, at 600°C, the excess molar Gibbs free energy of formation of the solid solutions is given by

$$G^{XS} = -28,280 X_{Au} X_{Cu} \text{ J}$$

Calculate the partial pressures of Au and Cu exerted by the solid solution of $X_{Cu} = 0.6$ at 600°C.
The saturated vapor pressure of solid copper is given by

$$\ln p^{\circ}_{Cu}(\text{atm}) = -\frac{40,920}{T} - 0.86\ln T + 21.67$$

and the saturated vapor pressure of solid gold is given by

$$\ln p^{\circ}_{Au}(\text{atm}) = -\frac{45,650}{T} - 0.306\ln T + 10.81$$

5. Al_2O_3, which melts at 2324 K, and Cr_2O_3, which melts at 2538 K form complete ranges of solid and liquid solutions. Assuming that $\Delta S^{\circ}_{m,Cr_2O_3} = \Delta S^{\circ}_{m,Al_2O_3}$, and that the solid and liquid solutions in the system Al_2O_3-Cr_2O_3 behave ideally, $\Delta H^{\circ}_{m(Al_2O_3)} = 107.5 kJ/mole$, calculate

1) The temperature at which equilibrium melting begins when an alloy of $X_{Al_2O_3} = 0.5$ is heated,

2) The composition of the melt which first forms,

3) The temperature at which equilibrium melting is complete,
4) The composition of last-formed solid.

6. A certain solid solution of A and B has a ΔG_{mix}^{XS} as follows:

$$\Delta G_{mix}^{XS} = \Delta G_{\mathrm{mix}} - \Delta G_{mix}^{id} = X_B X_A [a_1 X_A + a_2 X_B] + RT[X_A \ln X_A + X_B X_B]$$

Where:
a_1 = 12,500 J/mole
a_2 = 5,500 J/mole

a. Plot - ΔG_{mix}^{XS} at T = 500 and 700 K.

b. Sketch the T versus X B phase diagram of this alloy.
c. Determine the critical temperature and composition of this alloy.